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Journal of Vibration and Control
2022, Vol. 0(0) 1–20
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Wave-based in-plane vibration analysis of multiple coupled beam structures with arbitrary connection angle and elastic boundary restraints

Pengxin Tao, Yang Liu, Jingtao Du and Zhigang Liu

Abstract

Beam structures are widely used in various engineering branches, and the investigation on their dynamic behavior is of fundamental significance for the efficient design and vibration control of complex beam structures. In this work, a unified wave-based model for the in-plane vibration analysis of multiple coupled beams with arbitrary connection angle and boundary conditions is established, in which the vibrations within structures are described as wave propagation along structural elements, accompanying with wave reflection and transmission at some particular positions (such as the boundaries or structural joints). The reflection and transmission matrices corresponding to the incident waves propagating to the angular joint from various directions and the wave reflection matrix at the elastic boundary are derived. The natural and forced in-plane vibrations of multiple coupled beams can be analyzed concisely and systematically by assembling these wave reflection and transmission matrices. The accuracy of this approach is validated by comparing the dynamic characteristics obtained by the current approach with finite element method. Then, the influences of boundary stiffnesses and joint angle on wave reflection and transmission matrices are investigated. The results show that it is possible to know which wave type is dominant in structure at a certain frequency or angle, which shows certain significance for future work of vibration control and power flow analysis. The unified analysis model established in this paper can not only analyze the vibration characteristics of complex coupled beams but also reveal how waves are reflected and transmitted in the structure.

Keywords

multiple beam structures, wave analysis, connection angle, reflection and transmission matrices, boundary restraints

1. Introduction

As one of the basic structural elements in engineering, flexile beams have a wide range of applications in coupling systems and various fields, such as large bridges, planar frames, and various support brackets on ships. Due to the influence of external excitation, excessive beam vibration is usually encountered, which usually causes the structural fatigue and radiates noise. For this reason, a lot of research effort has been devoted to the dynamic characteristics of various beam-type structures.

Modal analysis method (Du, 2007; Li, 2000, 2002; Meirovitch, 1967; Pan, 1998) or wave propagation approach (Gao, 2021; Mead, 1975a, 1975b, 1986, 1996; Yong, 1992; Xiao, 2013; Wandowski, 2016) are usually adopted for the vibration analysis of structures. In comparison, the modal analysis method is a more mathematical concept and the structural response to any excitation force can be described as the superposition of all the eigenmodes. The wave propagation approach is a more physical concept and the structural response is the superposition of the several wave components. Although modal analysis method can be used to solve complex structural vibration problems, some microscopic vibration parameters such as vibration wavelength, reflected and transmitted waves cannot be obtained and investigated. Wave propagation approach is widely used because it is convenient for researchers and

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Received: 11 April 2022; accepted: 28 September 2022

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designers to understand and observe structural vibration at a more microscopic and physical level.

Scholars around the word have carried out a lot of researches on complex coupled beam structures by using wave propagation approach. Some scholars used matrix expressions coupled together to investigate the dynamic characteristics of complex structures based on wave propagation approach. For example, some scholars analyzed and investigated the overall dynamic characteristics of the complex structures by coupling the expressions of the four unknown coefficients of each single beam, including wave amplitude, displacement, shear force, and bending moment in a matrix formulation. Therefore, dynamic stiffness method (Banerjee, 1992, 2019), mobility approach (Gardonio, 2002), wave scattering method (Beale, 1995; Cai, 1991; Tomita, 2020), ray tracing method (Chouvion, 2010, 2011; Tang, 2015), and spectral element method (Lee, 2009; Samaratunga, 2014) based on wave propagation were proposed. But for these kinds of methods, once the structural model is too complex, the analysis difficulty will be greatly increased. Taking the mobility approach as an example, if the complex coupled beam structures have more than one connection point, it is necessary to construct a special norm to solve vibration problems, which makes the analysis and calculation difficult. Moreover, these methods based on wave propagation mostly borrow wave propagation as a tool to analyze the dynamic characteristics of structures, without considering how waves are reflected and transmitted at a microscopic level.

There are some other scholars using the wave propagation and attenuation characteristics to describe the vibration behavior of the structures. They analyzed the reflection and transmission of waves at discontinuities of beam structures based on wave propagation approach so that the structural vibration can be analyzed concisely and systematically through the superposition of various incident, reflected and transmitted waves. Desmond (1981) investigated the propagation characteristics of waves in rods and the reflection and transmission coefficients at a junction. Mace (1984) derived the wave propagation, reflection, and transmission coefficient matrices in Euler-Bernoulli beams, which can be used to simplify the vibration analysis at discontinuities of the beams. On the basis, Mei has made great contributions to the investigation of vibration characteristics of frame structures. In her series of works, she has investigated the dynamic properties of H- and T-shaped beams (2008), L-shaped beams (2012), and multi-story planner frames (2014). She reported the free and forced in-plane vibrations of Timoshenko beams and Euler-Bernoulli beams. Later, she expanded her investigation from in-plane vibration to out-of-plane vibration, as well as space frame structures (2019). Using the same train of thought, some structural dynamics problems have been solved and investigated in practical engineering applications.

In the previous work, most of the investigations and analyses are based on classical boundary conditions, and the case of general elastic boundary restraints is rarely considered. Moreover, the existing work is more limited to straight structural element connected perpendicularly, and the analysis of multiple coupled angular beam structures has rarely been addressed, while the multiple coupled angular beam structures are very common in reality. Therefore, the wave-based approach to the analysis of vibration of multiple coupled angular beam structures with elastic boundaries is more necessary and can be applied to a wider range of one-dimensional structures in practice.

The main objective of this work is to present an exact analytical method based on wave propagation, reflection, and transmission characteristics for investigating the vibration behavior of multiple coupled angular beam structures. In this respect, the work in this paper can be considered as a more general extension for the orthogonal joint structures. It should be noted that when the joint angle changes, the mass and moment of inertia of the joint will vary with the geometry of the joint and are difficult to represent in unified mathematical expressions. For the consistency of derivation of more complex coupled beam structures in future work, the mass and moment of inertia of the joint can be neglected to facilitate a unified analysis. This simplification has also been used by many scholars (Caglar, 2018; Kang, 2003; Huang, 2013; Tomita, 2020) to analyze complex waveguide structures. And the accuracy and precision are acceptable in practical engineering. In addition, the effects of boundary stiffness and joint angle on the reflection and transmission matrices are further analyzed in this paper. This is of positive significance for the subsequent research of vibration control and vibrational energy prediction.

The organization of this paper is as follows: In section 2, the equations of wave propagation are present. In section 3, formulations of the multiple coupled angular beams are derived. In section 4, free and forced vibration are analyzed, and numerical results are given. The effects of boundary stiffness and joint angle on reflection and transmission coefficients are investigated in section 5. Section 6 is the conclusion of this paper.

2. Equation of wave propagation

When considering the classical beam or rod related theories (Doyle, 1989; Meirovitch, 2000), the governing equations of motion for the transverse and longitudinal vibration of a uniform beam are expressed in the following equations, respectively:

$$EI\frac{\partial^4 y(x,t)}{\partial x^4} + m\frac{\partial^2 y(x,t)}{\partial t^4} = q(x,t) \tag{1}$$

$$m\frac{\partial^2 u(x,t)}{\partial t^4} - SE\frac{\partial^2 u(x,t)}{\partial x^2} = p(x,t) \tag{2}$$

In both equations, x is the position along the beam axis. E, S, I, and m are Young's modulus, area of the cross section, the area moment of inertia, and mass per unit length of the structure. $q(x, t)$ and $p(x, t)$ are externally transverse and longitudinal forces.

The shear force V, bending moment M, and longitudinal force F at any section of the beam are related to the longitudinal displacement $u(x, t)$, the transverse displacement $y(x, t)$, and the bending slope $\psi(x, t)$ by

$$F = SE\frac{\partial u}{\partial x} \tag{3}$$

$$V = -IE\frac{\partial^3 y}{\partial x^3} \tag{4}$$

$$M = IE\frac{\partial \psi}{\partial x} \tag{5}$$

where $\psi = \frac{\partial y}{\partial x}$ is based on the classical Euler-Bernoulli beam theory.

Using separation of variables, the solutions to the governing equations of beam and rod can be written as

$$y(x) = a_1^+ e^{-ik_1 x} + a_2^+ e^{-k_1 x} + a_1^- e^{ik_1 x} + a_2^- e^{k_1 x} \tag{6}$$

$$u(x) = c^+ e^{-ik_2 x} + c^- e^{ik_2 x} \tag{7}$$

The superscript + and – indicate positive and negative going waves.Where the bending wavenumber and the longitudinal wavenumber are

$$k_1 = \sqrt[4]{\rho S\omega^2/IE} \tag{8}$$

$$k_2 = \sqrt{\frac{\rho}{E}}\omega \tag{9}$$

here, ω is the frequency and ρ is the mass density.

Wave amplitudes at two points A and B on a uniform beam can be related using the propagation matrix. Considering the positive and negative going wave vectors at points A and B as $\mathbf{a}^+$, $\mathbf{a}^-$, $\mathbf{b}^+$, and $\mathbf{b}^-$, respectively. They are related by

$$\mathbf{a}^- = \mathbf{f}(x)\mathbf{b}^- \quad \mathbf{b}^+ = \mathbf{f}(x)\mathbf{a}^+ \tag{10}$$

where

$$\mathbf{f}(x) = \begin{bmatrix} e^{-ik_1 x} & 0 & 0 \\ 0 & e^{-k_1 x} & 0 \\ 0 & 0 & e^{-ik_2 x} \end{bmatrix} \tag{11}$$

is the propagation matrix for a distance x.

And

$$\mathbf{a}^+ = \begin{bmatrix} a_1^+ \\ a_2^+ \\ c^+ \end{bmatrix} \quad \mathbf{a}^- = \begin{bmatrix} a_1^- \\ a_2^- \\ c^- \end{bmatrix} \quad \mathbf{b}^+ = \begin{bmatrix} b_1^+ \\ b_2^+ \\ d^+ \end{bmatrix} \quad \mathbf{b}^- = \begin{bmatrix} b_1^- \\ b_2^- \\ d^- \end{bmatrix} \tag{12}$$

3. Formulation of the multiple coupled angular beams

Waves can be reflected and transmitted at the discontinuities in the structures. In this section, the reflection and transmission matrices of the multiple coupled angular beams will be considered and derived. A schematic diagram of the model is shown in Figure 1. For the convenience of subsequent description, each part of the multiple coupled angular beams is denoted by beam 1, beam 2, and beam 3, respectively. At the boundary of the beam, the general boundary conditions can be represented by translational spring, rotational, and longitudinal springs. Different spring stiffness can represent different boundary conditions.

Due to the reflection at the boundary, the set of incident waves $\mathbf{a}^+$ give rise to reflected waves $\mathbf{a}^+ = \mathbf{r}\mathbf{a}^-$. These are schematically illustrated in Figure 2.

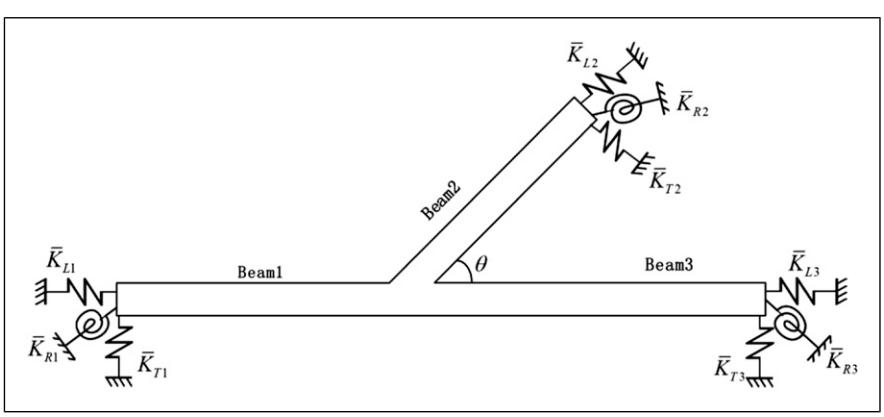

Figure 1. Model illustration of the multiple coupled angular beams with elastic boundary conditions.

According to equilibrium at the boundary:

$$\begin{cases} V(x) = \overline{K}_T y \\ M(x) = \overline{K}_T \dfrac{\partial y}{\partial x} \\ F(x) = \overline{K}_L u \end{cases} \tag{13}$$

The reflection matrix of elastic boundary $\mathbf{r}_e$ is obtained by substituting equations (3) ~ (7) into equations (13).

$$\mathbf{r}_e = \begin{bmatrix} 1 - iK_T & i(1-K_T) & 0 \\ -1 + iK_R & 1 + K_R & 0 \\ 0 & 0 & i + K_L \end{bmatrix}^{-1} \times \begin{bmatrix} 1 + iK_T & i(1+K_T) & 0 \\ 1 + iK_R & -1 + K_R & 0 \\ 0 & 0 & i - K_L \end{bmatrix} \tag{14}$$

where

$$K_T = \frac{\overline{K}_T}{IEk_1^3} \quad K_R = \frac{\overline{K}_R}{IEk_1} \quad K_L = \frac{\overline{K}_L}{SEk_2} \tag{15}$$

are the dimensionless stiffnesses.

Classical boundary conditions can be obtained by setting K_T, K_R, and K_L to zero or infinity, accordingly. The reflection matrices at clamped boundary $\mathbf{r}_c$, free boundary $\mathbf{r}_f$, and simply-supported boundary $\mathbf{r}_s$ are given by the following equations, respectively.

$$\mathbf{r}_c = \begin{bmatrix} -i & -1-i & 0 \\ -1+i & i & 0 \\ 0 & 0 & -1 \end{bmatrix} \tag{16}$$

$$\mathbf{r}_f = \begin{bmatrix} -i & 1+i & 0 \\ 1-i & i & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{17}$$

$$\mathbf{r}_s = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \tag{18}$$

For deriving the reflection and transmission matrices at the angular T joint, beam 1, beam 2, and beam 3 are supposed to be infinite. Figure 3 is a schematic illustration of the stress diagram of the free body and the equations of motion at the joint in Figure 3 are

$$\begin{aligned} -M_1 + M_2 + M_3 &= 0 \\ -V_1 + V_3 + F_2\sin\theta + V_2\cos\theta &= 0 \\ -F_1 + F_3 + F_2\cos\theta - V_2\sin\theta &= 0 \end{aligned} \tag{19}$$

where the subscripts 1, 2, and 3 denote parameters related to beam 1, beam 2, and beam 3, respectively.

There are three cases of wave reflection and transmission when waves are incident from different beam elements, as

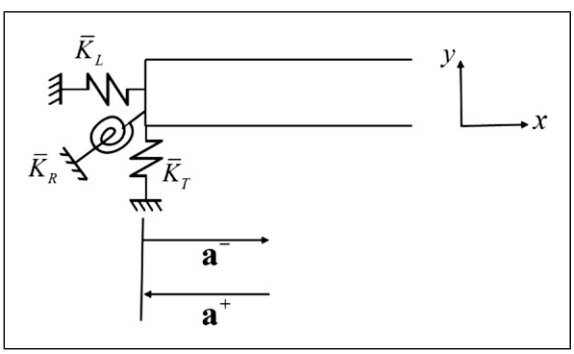

Figure 2. Schematic illustration of reflection at elastic boundary.

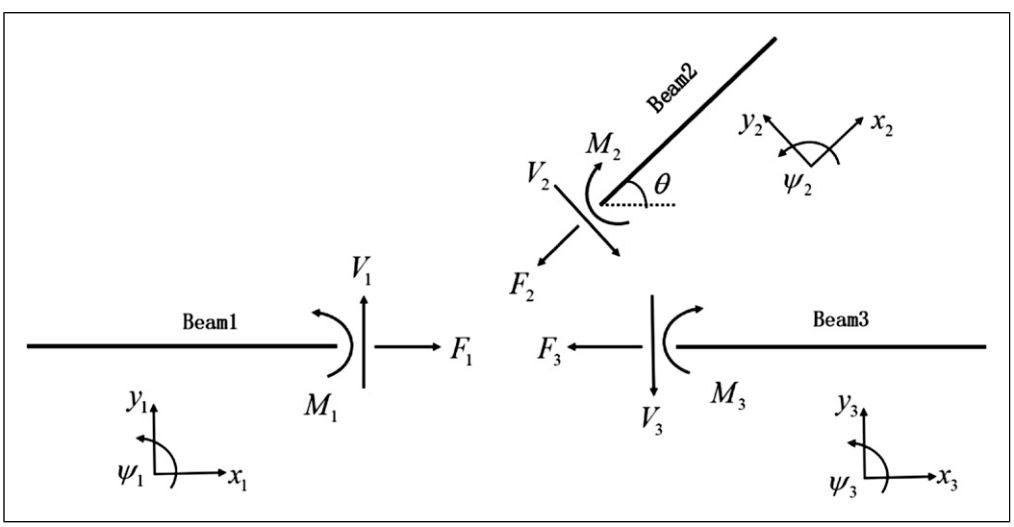

Figure 3. Free body diagram at angular joint.

shown in Figure 4. First, consider the case in Figure 4(a). When the waves $\mathbf{a}^+$ are incident from beam 1, reflected waves $\mathbf{a}^-$ and transmitted waves $\mathbf{b}^+$ and $\mathbf{e}^+$ will be generated at the joint, respectively. They are associated with incident waves by reflection and transmission matrices.

$$\mathbf{a}^- = \mathbf{r}_{11}\mathbf{a}^+ \quad \mathbf{b}^+ = \mathbf{t}_{12}\mathbf{a}^+ \quad \mathbf{e}^+ = \mathbf{t}_{13}\mathbf{a}^+ \tag{20}$$

where wave vectors $\mathbf{a}^\pm$ and $\mathbf{b}^\pm$ are as defined in equation (12), and

$$\mathbf{e}^\pm = \begin{bmatrix} e_1^\pm \\ e_2^\pm \\ g^\pm \end{bmatrix} \tag{21}$$

Because there are incident and reflected waves in beam 1 and only transmitted waves in beam 2 and beam 3, the transverse and longitudinal displacements are given by

$$\begin{aligned} y_1 &= a_1^+ e^{-ik_1x_1} + a_2^+ e^{-k_1x_1} + a_1^- e^{ik_1x_1} + a_2^- e^{k_1x_1} \\ y_2 &= b_1^+ e^{-ik_1x_2} + b_2^+ e^{-k_1x_2} \\ y_3 &= e_1^+ e^{-ik_1x_3} + e_2^+ e^{-k_1x_3} \end{aligned} \tag{22}$$

$$u_1 = c^+ e^{-ik_2x_1} + c^- e^{ik_2x_1} \quad u_2 = d^+ e^{-ik_2x_2} \quad u_3 = g^+ e^{-ik_2x_3} \tag{23}$$

$$\begin{aligned} \psi_1 &= -ik_1 a_1^+ e^{-ik_1x_1} - k_1 a_2^+ e^{-k_1x_1} + ik_1 a_1^- e^{ik_1x_1} + k_1 a_2^- e^{k_1x_1} \\ \psi_2 &= -ik_1 b_1^+ e^{-ik_1x_2} - k_1 b_2^+ e^{-k_1x_2} \\ \psi_3 &= -ik_1 e_1^+ e^{-ik_1x_3} - k_1 e_2^+ e^{-k_1x_3} \end{aligned} \tag{24}$$

The continuity conditions equations at the joint can be divided into two groups, one is about beam 1 and beam 2 and the other is about beam 1 and beam 3.

$$\begin{aligned} u_1 &= u_2\cos\theta - y_2\sin\theta \\ y_1 &= u_2\sin\theta + y_2\cos\theta \\ \psi_1 &= \psi_2 \end{aligned} \tag{25}$$

$$\begin{aligned} u_1 &= u_3 \\ y_1 &= y_3 \\ \psi_1 &= \psi_3 \end{aligned} \tag{26}$$

From equations (3) ~ (5), (19) and (22), and (23) and (24), the equilibrium conditions equations can be expressed in matrix form as follow:

$$\mathbf{N}_1\mathbf{a}^+ + \mathbf{N}_2\mathbf{a}^- + \mathrm{N}_3\mathbf{b}^+ + \mathbf{N}_4\mathbf{e}^+ = 0 \tag{27}$$

where

$$\mathbf{N}_1 = \begin{bmatrix} ik_1^3 & -k_1^3 & 0 \\ 0 & 0 & ik_2 \\ 1 & -1 & 0 \end{bmatrix} \quad \mathbf{N}_2 = \begin{bmatrix} -ik_1^3 & k_1^3 & 0 \\ 0 & 0 & -ik_2 \\ 1 & -1 & 0 \end{bmatrix}$$

$$\mathbf{N}_3 = \begin{bmatrix} -ik_1^3\cos\theta & k_1^3\cos\theta & -i\Omega k_2\sin\theta \\ i\frac{1}{\Omega}k_1^3\sin\theta & -\frac{1}{\Omega}k_1^3\sin\theta & -ik_2\cos\theta \\ -1 & 1 & 0 \end{bmatrix}$$

$$\mathbf{N}_4 = \begin{bmatrix} -ik_1^3 & k_1^3 & 0 \\ 0 & 0 & -ik_2 \\ -1 & 1 & 0 \end{bmatrix}$$

here, $\Omega = \frac{S}{I}$ is defined for simplicity of matrix expression.

From equations (3) ~ (5) and (22), (23), (24), (25), and (26), the continuity conditions equations between beam 1 and beam 2 and between beam 1 and beam 3 can be expressed in matrix form as follows:

$$\mathbf{N}_5\mathbf{a}^+ + \mathbf{N}_6\mathbf{a}^- + \mathrm{N}_7\mathbf{b}^+ = 0 \tag{28}$$

$$\mathbf{N}_8\mathbf{a}^+ + \mathbf{N}_9\mathbf{a}^- + \mathrm{N}_{10}\mathbf{e}^+ = 0 \tag{29}$$

where

$$\mathbf{N}_5 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ -i & -1 & 0 \end{bmatrix}$$

$$\mathbf{N}_6 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ i & 1 & 0 \end{bmatrix} \quad \mathbf{N}_7 = \begin{bmatrix} \sin\theta & \sin\theta & -\cos\theta \\ -\cos\theta & -\cos\theta & -\sin\theta \\ i & 1 & 0 \end{bmatrix}$$

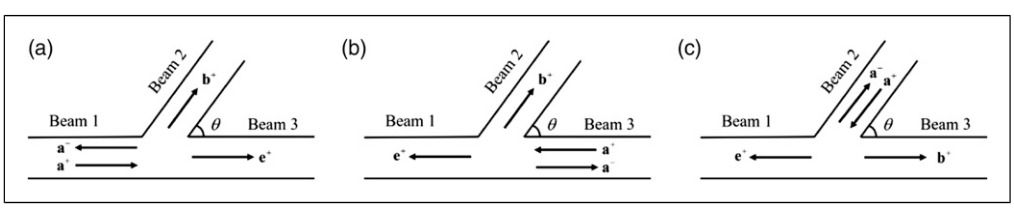

Figure 4. Schematic illustration of wave reflection and transmission at the joint. (a) incident wave from beam 1; (b) incident wave from beam 3; (c) incident wave from beam 2.

$$\mathbf{N}_8 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ -i & -1 & 0 \end{bmatrix} \mathbf{N}_9 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ i & 1 & 0 \end{bmatrix}$$

$$\mathbf{N}_{10} = \begin{bmatrix} 0 & 0 & -1 \\ -1 & -1 & 0 \\ i & 1 & 0 \end{bmatrix}$$

From equations (20) and (27) ~ (28), a system of equations about reflection and transmission matrices $\mathbf{r}_{11}$, $\mathbf{t}_{12}$, and $\mathbf{t}_{13}$ can be obtained.

$$\begin{aligned} \mathbf{N}_1 + \mathbf{N}_2\mathbf{r}_{11} + \mathrm{N}_3\mathbf{t}_{12} + \mathbf{N}_4\mathbf{t}_{13} &= 0 \\ \mathbf{N}_5 + \mathbf{N}_6\mathbf{r}_{11} + \mathrm{N}_7\mathbf{t}_{12} &= 0 \\ \mathbf{N}_8 + \mathbf{N}_9\mathbf{r}_{11} + \mathrm{N}_{10}\mathbf{t}_{13} &= 0 \end{aligned} \tag{30}$$

The reflection and transmission matrices $\mathbf{r}_{11}$, $\mathbf{t}_{12}$, and $\mathbf{t}_{13}$ can be obtained from solving equations (30).

$$\begin{aligned} \mathbf{r}_{11} &= \left(\mathbf{N}_2 - \mathbf{N}_3\mathbf{N}_7^{-1}\mathbf{N}_6 - \mathbf{N}_4\mathbf{N}_{10}^{-1}\mathbf{N}_9\right)^{-1} \\ &\quad \left(-\mathbf{N}_1 + \mathbf{N}_3\mathbf{N}_7^{-1}\mathbf{N}_5 + \mathbf{N}_4\mathbf{N}_{10}^{-1}\mathbf{N}_8\right) \\ \mathbf{t}_{12} &= -\mathbf{N}_7^{-1}\mathbf{N}_5 - \mathbf{N}_7^{-1}\mathbf{N}_6\mathbf{r}_{11} \\ \mathbf{t}_{13} &= -\mathbf{N}_{10}^{-1}\mathbf{N}_8 - \mathbf{N}_{10}^{-1}\mathbf{N}_9\mathbf{r}_{11} \end{aligned} \tag{31}$$

When the waves are incident from beam 3, as shown in Figure 4(b). Incident waves $\mathbf{a}^+$ give rise to reflected waves $\mathbf{a}^-$ and transmitted waves $\mathbf{b}^+$ and $\mathbf{e}^+$. The relationship between them can be expressed by reflection and transmission matrices in the following equations.

$$\mathbf{a}^- = \mathbf{r}_{33}\mathbf{a}^+ \quad \mathbf{b}^+ = \mathbf{t}_{32}\mathbf{a}^+ \quad \mathbf{e}^+ = \mathbf{t}_{31}\mathbf{a}^+ \tag{32}$$

The two continuity conditions related equations at the joint between beam 3 and beam 2 and between beam 3 and beam 1 are

$$\begin{aligned} u_3 &= u_2\cos\theta - y_2\sin\theta \\ y_3 &= u_2\sin\theta + y_2\cos\theta \\ \psi_3 &= \psi_2 \end{aligned} \tag{33}$$

$$\begin{aligned} u_1 &= u_3 \\ y_1 &= y_3 \\ \psi_1 &= \psi_3 \end{aligned} \tag{34}$$

Taking a similar procedure, the two continuity conditions equations (33), (34) and the equilibrium conditions equations (19) can be expressed in matrix form as follows:

$$\mathbf{M}_1\mathbf{a}^+ + \mathbf{M}_2\mathbf{a}^- + \mathbf{M}_3\mathbf{b}^+ + \mathbf{M}_4\mathbf{e}^+ = 0 \tag{35}$$

$$\mathbf{M}_5\mathbf{a}^+ + \mathbf{M}_6\mathbf{a}^- + \mathbf{M}_7\mathbf{b}^+ = 0 \tag{36}$$

$$\mathbf{M}_8\mathbf{a}^+ + \mathbf{M}_9\mathbf{a}^- + \mathbf{M}_{10}\mathbf{e}^+ = 0 \tag{37}$$

where,

$$\mathbf{M}_1 = \begin{bmatrix} ik_1^3 & -k_1^3 & 0 \\ 0 & 0 & ik_2 \\ -1 & 1 & 0 \end{bmatrix} \mathbf{M}_2 = \begin{bmatrix} -ik_1^3 & k_1^3 & 0 \\ 0 & 0 & -ik_2 \\ -1 & 1 & 0 \end{bmatrix}$$

$$\mathbf{M}_3 = \begin{bmatrix} -ik_1^3\cos\theta & k_1^3\cos\theta & -i\Omega k_2\sin\theta \\ i\dfrac{1}{\Omega}k_1^3\sin\theta & -\dfrac{1}{\Omega}k_1^3\sin\theta & -ik_2\cos\theta \\ -1 & 1 & 0 \end{bmatrix}$$

$$\mathbf{M}_4 = \begin{bmatrix} -ik_1^3 & k_1^3 & 0 \\ 0 & 0 & -ik_2 \\ 1 & -1 & 0 \end{bmatrix} \mathbf{M}_5 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ i & 1 & 0 \end{bmatrix}$$

$$\mathbf{M}_6 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ -i & -1 & 0 \end{bmatrix} \mathbf{M}_7 = \begin{bmatrix} \sin\theta & \sin\theta & -\cos\theta \\ -\cos\theta & -\cos\theta & -\sin\theta \\ i & 1 & 0 \end{bmatrix}$$

$$\mathbf{M}_8 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ i & 1 & 0 \end{bmatrix} \mathbf{M}_9 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ -i & -1 & 0 \end{bmatrix}$$

$$\mathbf{M}_{10} = \begin{bmatrix} 0 & 0 & -1 \\ -1 & -1 & 0 \\ -i & -1 & 0 \end{bmatrix}$$

From equations (32) and (35) ~ (37),

$$\begin{aligned} \mathbf{M}_1 + \mathbf{M}_2\mathbf{r}_{33} + \mathbf{M}_3\mathbf{t}_{32} + \mathbf{M}_4\mathbf{t}_{31} &= 0 \\ \mathbf{M}_5 + \mathbf{M}_6\mathbf{r}_{33} + \mathbf{M}_7\mathbf{t}_{32} &= 0 \\ \mathbf{M}_8 + \mathbf{M}_9\mathbf{r}_{33} + \mathbf{M}_{10}\mathbf{t}_{31} &= 0 \end{aligned} \tag{38}$$

The reflection and transmission matrices $\mathbf{r}_{33}$, $\mathbf{t}_{32}$, and $\mathbf{t}_{31}$ can be obtained from solving equations (38).

$$\begin{aligned} \mathbf{r}_{33} &= \left(\mathbf{M}_2 - \mathbf{M}_3\mathbf{M}_7^{-1}\mathbf{M}_6 - \mathbf{M}_4\mathbf{M}_{10}^{-1}\mathbf{M}_9\right)^{-1} \\ &\quad \left(-\mathbf{M}_1 + \mathbf{M}_3\mathbf{M}_7^{-1}\mathbf{M}_5 + \mathbf{M}_4\mathbf{M}_{10}^{-1}\mathbf{M}_8\right) \\ \mathbf{t}_{32} &= -\mathbf{M}_7^{-1}\mathbf{M}_5 - \mathbf{M}_7^{-1}\mathbf{M}_6\mathbf{r}_{33} \\ \mathbf{t}_{31} &= -\mathbf{M}_{10}^{-1}\mathbf{M}_8 - \mathbf{M}_{10}^{-1}\mathbf{M}_9\mathbf{r}_{33} \end{aligned} \tag{39}$$

Finally, consider the case that waves incident from beam 2 shown in Figure 4(c). The relationship between incident, reflected, and transmitted waves can be expressed by reflection and transmission matrices in the following equations.

$$\mathbf{a}^- = \mathbf{r}_{22}\mathbf{a}^+ \quad \mathbf{b}^+ = \mathbf{t}_{23}\mathbf{a}^+ \quad \mathbf{e}^+ = \mathbf{t}_{21}\mathbf{a}^+ \tag{40}$$

The two continuity conditions related equations at the joint are

$$\begin{aligned} u_1 &= u_2\cos\theta - y_2\sin\theta \\ y_1 &= u_2\sin\theta + y_2\cos\theta \\ \psi_1 &= \psi_2 \end{aligned} \tag{41}$$

$$\begin{aligned} u_3 &= u_2\cos\theta - y_2\sin\theta \\ y_3 &= u_2\sin\theta + y_2\cos\theta \\ \psi_2 &= \psi_3 \end{aligned} \tag{42}$$

Again, taking a similar procedure, the two continuity conditions equations (41), (42) and the equilibrium conditions equations (19) can be expressed in matrix form as follows:

$$\mathbf{P}_1\mathbf{a}^+ + \mathbf{P}_2\mathbf{a}^- + \mathbf{P}_3\mathbf{b}^+ + \mathbf{P}_4\mathbf{e}^+ = 0 \tag{43}$$

$$\mathbf{P}_5\mathbf{a}^+ + \mathbf{P}_6\mathbf{a}^- + \mathbf{P}_7\mathbf{e}^+ = 0 \tag{44}$$

$$\mathbf{P}_8\mathbf{a}^+ + \mathbf{P}_9\mathbf{a}^- + \mathbf{P}_{10}\mathbf{b}^+ = 0 \tag{45}$$

where,

$$\mathbf{P}_1 = \begin{bmatrix} ik_1^3\cos\theta & -k_1^3\cos\theta & i\Omega k_2\sin\theta \\ -i\frac{1}{\Omega}k_1^3\sin\theta & \frac{1}{\Omega}k_1^3\sin\theta & ik_2\cos\theta \\ -1 & 1 & 0 \end{bmatrix}$$

$$\mathbf{P}_2 = \begin{bmatrix} -ik_1^3\cos\theta & k_1^3\cos\theta & -i\Omega k_2\sin\theta \\ i\frac{1}{\Omega}k_1^3\sin\theta & -\frac{1}{\Omega}k_1^3\sin\theta & -ik_2\cos\theta \\ -1 & 1 & 0 \end{bmatrix}$$

$$\mathbf{P}_3 = \begin{bmatrix} -ik_1^3 & k_1^3 & 0 \\ 0 & 0 & -ik_2 \\ -1 & 1 & 0 \end{bmatrix} \mathbf{P}_4 = \begin{bmatrix} -ik_1^3 & k_1^3 & 0 \\ 0 & 0 & -ik_2 \\ 1 & -1 & 0 \end{bmatrix}$$

$$\mathbf{P}_5 = \begin{bmatrix} \sin\theta & \sin\theta & -\cos\theta \\ \cos\theta & \cos\theta & \sin\theta \\ i & 1 & 0 \end{bmatrix} \mathbf{P}_6 = \begin{bmatrix} \sin\theta & \sin\theta & -\cos\theta \\ \cos\theta & \cos\theta & \sin\theta \\ -i & -1 & 0 \end{bmatrix}$$

$$\mathbf{P}_7 = \begin{bmatrix} 0 & 0 & 1 \\ -1 & -1 & 0 \\ -i & -1 & 0 \end{bmatrix} \mathbf{P}_8 = \begin{bmatrix} \sin\theta & \sin\theta & -\cos\theta \\ \cos\theta & \cos\theta & \sin\theta \\ i & 1 & 0 \end{bmatrix}$$

$$\mathbf{P}_9 = \begin{bmatrix} \sin\theta & \sin\theta & -\cos\theta \\ \cos\theta & \cos\theta & \sin\theta \\ -i & -1 & 0 \end{bmatrix}$$

$$\mathbf{P}_{10} = \begin{bmatrix} 0 & 0 & 1 \\ -1 & -1 & 0 \\ i & 1 & 0 \end{bmatrix}$$

From equations (40) and (43) ~ (45),

$$\begin{aligned} \mathbf{P}_1 + \mathbf{P}_2\mathbf{r}_{22} + \mathbf{P}_3\mathbf{t}_{23} + \mathbf{P}_4\mathbf{t}_{21} &= 0 \\ \mathbf{P}_5 + \mathbf{P}_6\mathbf{r}_{22} + \mathbf{P}_7\mathbf{t}_{21} &= 0 \\ \mathbf{P}_8 + \mathbf{P}_9\mathbf{r}_{22} + \mathbf{P}_{10}\mathbf{t}_{23} &= 0 \end{aligned} \tag{46}$$

The reflection and transmission matrices $\mathbf{r}_{22}$, $\mathbf{t}_{23}$, and $\mathbf{t}_{21}$ can be obtained from solving equations (46).

$$\begin{aligned} \mathbf{r}_{22} &= \left(\mathbf{P}_2 - \mathbf{P}_3\mathbf{P}_{10}^{-1}\mathbf{P}_9 - \mathbf{P}_4\mathbf{P}_7^{-1}\mathbf{P}_6\right)^{-1} \\ &\quad \left(-\mathbf{P}_1 + \mathbf{P}_3\mathbf{P}_{10}^{-1}\mathbf{P}_8 + \mathbf{P}_4\mathbf{P}_7^{-1}\mathbf{P}_5\right) \\ \mathbf{t}_{21} &= -\mathbf{P}_7^{-1}\mathbf{P}_5 - \mathbf{P}_7^{-1}\mathbf{P}_6\mathbf{r}_{22} \\ \mathbf{t}_{23} &= -\mathbf{P}_{10}^{-1}\mathbf{P}_8 - \mathbf{P}_{10}^{-1}\mathbf{P}_9\mathbf{r}_{22} \end{aligned} \tag{47}$$

4. Vibration analysis and numerical results

In the above sections, the reflection and transmitted matrices generated at the joint when transverse and longitudinal waves incident from different beam elements are derived. The vibration analysis of complex beam structures can be carried out concisely and systematically by using the combination of these matrices. In this section, the free and forced vibration of multiple coupled angular beam structures (as shown in Figure 1) are analyzed using this exact analytical method. Some numerical results are given and discussed. (The effects of damping are neglected in these results)

4.1. Free vibration

Using the present method, the free vibration wave analysis diagram of the multiple coupled angular beams (as shown in Figure 1) is shown in Figure 5. The discontinuities of the structure include one joint C and three boundaries A, B, and D. Incident waves give rise to only reflected waves at the boundary, but also transmitted waves at the joint. Figure 5 shows the final superposition of the waves in the structure. The reflection and transmission relations of the waves at these discontinuities are shown as below

At the joint C, one will have

$$\begin{aligned} \mathbf{c}_3^+ &= \mathbf{r}_{33}\mathbf{c}_3^- + \mathbf{t}_{13}\mathbf{c}_1^+ + \mathbf{t}_{23}\mathbf{c}_2^- \\ \mathbf{c}_1^- &= \mathbf{r}_{11}\mathbf{c}_1^+ + \mathbf{t}_{31}\mathbf{c}_3^- + \mathbf{t}_{21}\mathbf{c}_2^- \\ \mathbf{c}_2^+ &= \mathbf{r}_{22}\mathbf{c}_2^- + \mathbf{t}_{12}\mathbf{c}_1^+ + \mathbf{t}_{32}\mathbf{c}_3^- \end{aligned} \tag{48}$$

At boundaries A, B, and D, the relationship can be obtained

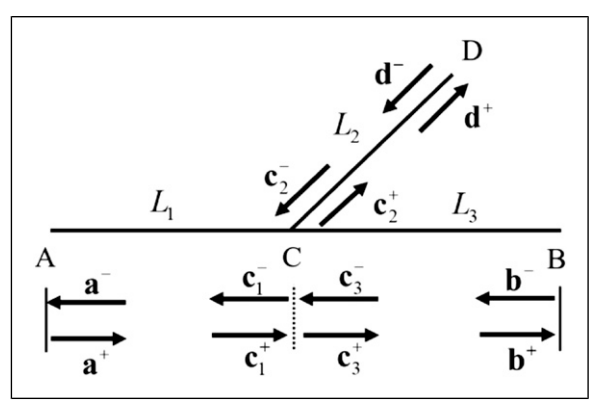

Figure 5. Schematic illustration of free vibration wave analysis.

$$\begin{aligned} \mathbf{a}^+ &= \mathbf{r}_A\mathbf{a}^- \\ \mathbf{b}^- &= \mathbf{r}_B\mathbf{b}^+ \\ \mathbf{d}^- &= \mathbf{r}_D\mathbf{d}^+ \end{aligned} \tag{49}$$

The boundary reflection matrices $\mathbf{r}_A$, $\mathbf{r}_B$, and $\mathbf{r}_D$ can be obtained from equation (14) according to different spring stiffnesses at the boundary.

Using propagation matrix equation (11), the propagation relations along uniform beam elements can be obtained.

Along AC

$$\mathbf{c}_1^+ = \mathbf{f}(L_1)\mathbf{a}^+ \quad \mathbf{a}^- = \mathbf{f}(L_1)\mathbf{c}_1^- \tag{50}$$

Along CB

$$\mathbf{b}^+ = \mathbf{f}(L_3)\mathbf{c}_3^+ \quad \mathbf{c}_3^- = \mathbf{f}(L_3)\mathbf{b}^- \tag{51}$$

Along CD

$$\mathbf{d}^+ = \mathbf{f}(L_2)\mathbf{c}_2^+ \quad \mathbf{c}_2^- = \mathbf{f}(L_2)\mathbf{d}^- \tag{52}$$

Writing these equations in matrix form, one can get

$$\mathbf{Az} = \mathbf{0} \tag{53}$$

where $\mathbf{A}$ is a 36 × 36 coefficient matrix. $\mathbf{z}$ is a 36 × 1 wave component vector. Setting the determinant of the coefficient matrix $\mathbf{A}$ to zero can obtain the natural frequencies of the structure. Mode shapes can be obtained from the corresponding eigenvectors.

The physical parameters of the beam structure are $L_1 = L_2 = L_3 = 0.5$ m, Young's modulus $E = 2.06 \times 10^{11}$ N/m^2, and mass density $\rho = 7800$ kg/m^3. The width and thickness of the beam structures are both 1.5×10^{-2} m. For the purpose of comparison and validation of the accuracy of the method, the first six natural frequencies and FEM simulation results of the multiple coupled angular beams with different angles are listed in Table 1. Boundary A is clamped and B and D are both free. The FEM model is a 2D geometric model established by using beam model in COMSOL Multiphysics software. The complete mesh contains 150 elements and the length of the element is 0.01 m, which satisfies the calculation accuracy. It can be seen from Table 1 that the natural frequencies obtained by this method are in good agreement with the FEM simulation results.

At the same time, Table 2 also gives the first six natural frequencies of the multiple coupled angular beams with different angles under elastic boundary conditions, which is rarely involved by most researchers before. Boundary A is elastic, which the spring stiffnesses are $K_{T1} = K_{L1} = 1 \times 10^4$ N/m and $K_{R1} = 1 \times 10^4$ Nm/rad. Boundary B and D are both free.

Mode shapes of the first four modes of the angular beam structure are shown in Figure 6, which the angle is 60° and the boundary A is clamped, B and D are both free.

Meanwhile, the first four modes of the 90° angular beam structure are shown in Figure 7. The boundary conditions are the same as before.

4.2. Forced vibration

An external excitation is applied at some point on a uniform beam. The excitation point is a new discontinuity of the

Table 1. Natural frequencies of the multiple coupled angular beams (clamped-free-free) (Hz).

	30°		45°		60°		75°		90°	
	FEM	Present	FEM	Present	FEM	Present	FEM	Present	FEM	Present
1st	9.18	9.32	9.35	9.38	9.59	9.65	9.90	9.79	10.26	10.32
2nd	49.72	49.59	49.27	49.19	47.89	47.92	45.39	45.33	42.55	42.59
3rd	67.22	67.22	62.80	62.66	59.76	59.90	58.98	58.91	59.96	59.90
4th	221.48	221.63	220.34	220.39	219.35	219.30	218.68	218.67	218.36	218.30
5th	312.01	312.15	311.89	311.76	311.83	311.81	311.91	311.73	312.20	312.15
6th	393.69	393.68	374.83	374.87	362.00	362.06	354.90	354.87	352.51	352.41

Table 2. Natural frequencies of the multiple coupled angular beams (elastic-free-free) (Hz).

	30°	45°	60°	75°	90°
1st	6.39	6.39	6.69	6.69	6.99
2nd	9.68	9.68	9.68	9.68	9.68
3rd	22.85	22.25	21.65	21.05	20.16
4th	50.38	51.28	52.78	53.98	55.77
5th	96.78	93.78	90.49	87.80	86.00
6th	249.72	246.02	243.73	240.74	238.34

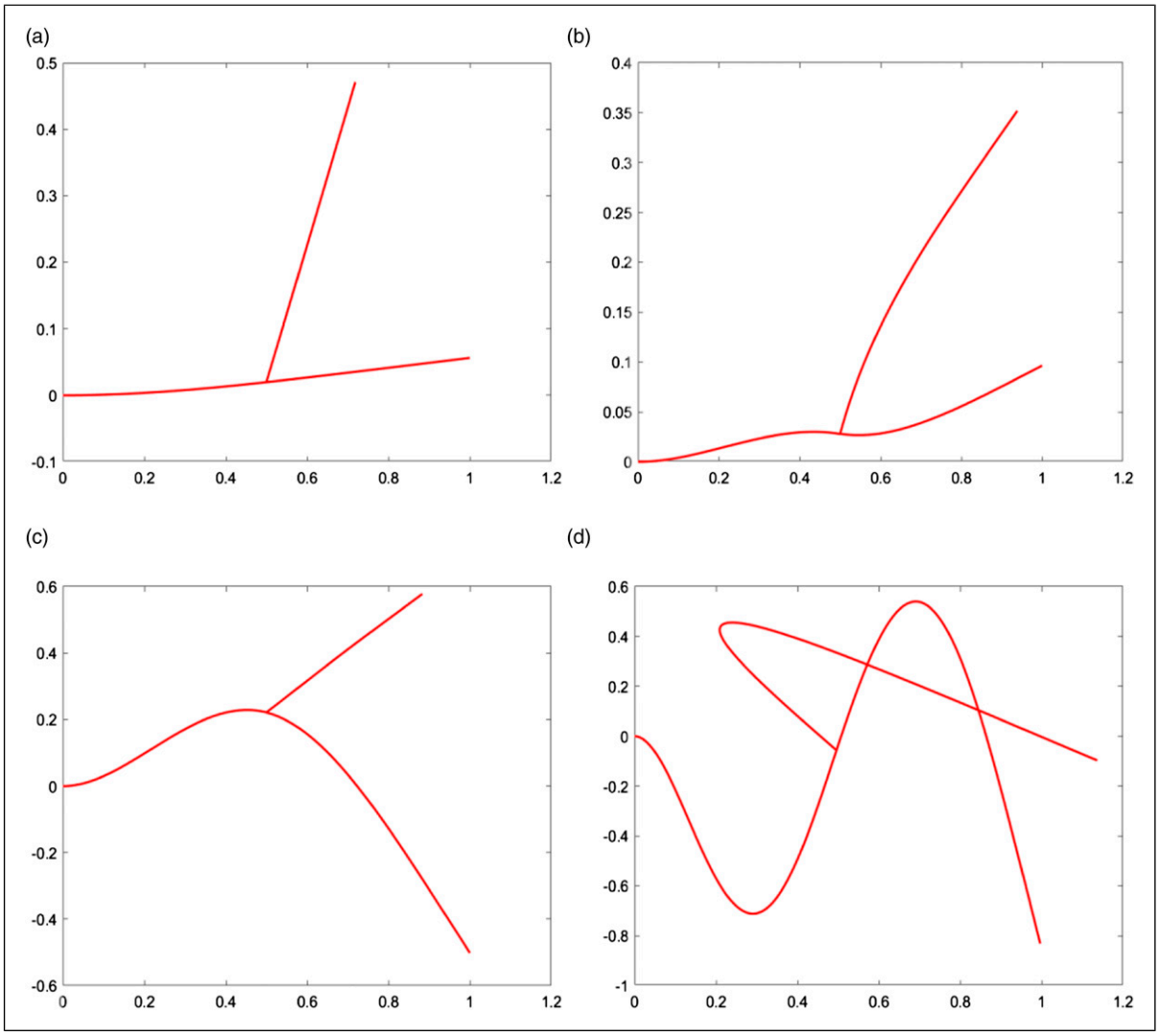

Figure 6. Mode shapes of the 60° multiple coupled angular beams (clamped-free-free). (a) the first mode shape, (b) the second mode shape, (c) the third mode shape, (d) the fourth mode shape.

beam, which will give rise to waves $\mathbf{a}^{\pm}$ and $\mathbf{b}^{\pm}$ in different directions on both sides of the excitation point. The relations between the external forces and the waves are expressed by the following equations

$$\mathbf{b}^{+} - \mathbf{a}^{+} = \mathbf{q} + \mathbf{m} - \mathbf{f} \quad \mathbf{b}^{-} - \mathbf{a}^{-} = -\mathbf{q} + \mathbf{m} + \mathbf{f} \tag{54}$$

where

$$\mathbf{q} = \begin{bmatrix} -i \\ -1 \\ 0 \end{bmatrix} \frac{Q}{4EIk_1^3} \quad \mathbf{m} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \frac{M}{4EIk_1^2} \quad \mathbf{f} = \begin{bmatrix} 0 \\ 0 \\ i \end{bmatrix} \frac{F}{2ESk_2} \tag{55}$$

here, Q, M, and F represent the point transverse force, bending moment, and longitudinal force, respectively.

Figure 8 shows the schematic illustration of forced vibration wave analysis.

At the external forces applied point G

$$\mathbf{g}_{12}^{+} - \mathbf{g}_{11}^{+} = \mathbf{q} + \mathbf{m} - \mathbf{f} \quad \mathbf{g}_{12}^{-} - \mathbf{g}_{11}^{-} = -\mathbf{q} + \mathbf{m} + \mathbf{f} \tag{56}$$

Along AC

$$\begin{aligned} \mathbf{g}_{11}^{+} &= \mathbf{f}(L_{11})\mathbf{a}^{+} \quad \mathbf{a}^{-} = \mathbf{f}(L_{11})\mathbf{g}_{11}^{-} \\ \mathbf{c}_{1}^{+} &= \mathbf{f}(L_{12})\mathbf{g}_{12}^{+} \quad \mathbf{g}_{12}^{-} = \mathbf{f}(L_{12})\mathbf{c}_{1}^{-} \end{aligned} \tag{57}$$

Writing these equations in matrix form gives

$$\mathbf{A}_f \mathbf{z}_f = \mathbf{F} \tag{58}$$

where $\mathbf{A}_f$ is a 48 × 48 coefficient matrix, $\mathbf{z}_f$ is a 48 × 1 wave component vector, and $\mathbf{F}$ is a 48 × 1 vector describing the externally applied forces and moments.

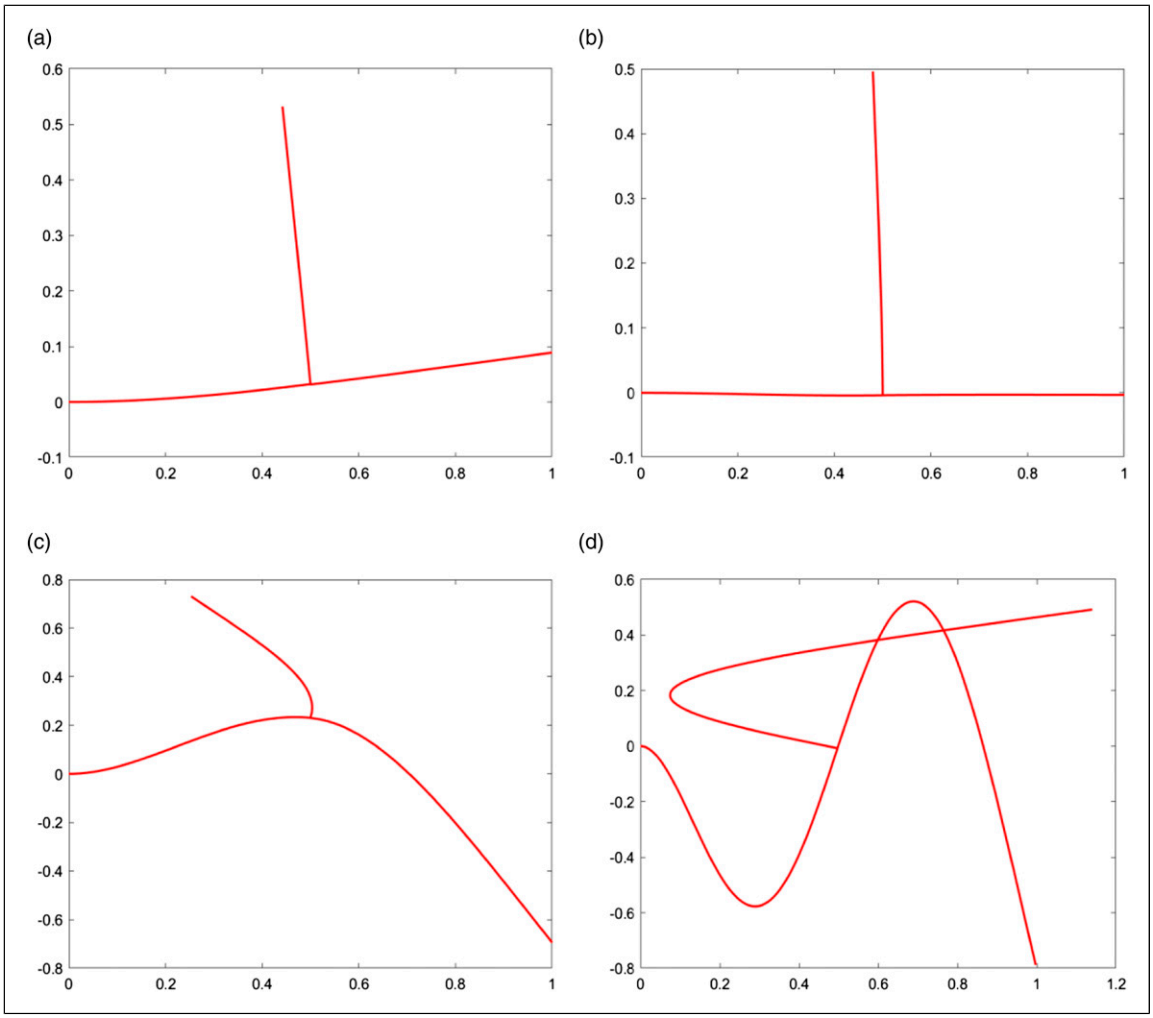

Figure 7. Mode shapes of the 90° multiple coupled angular beams (clamped-free-free). (a) the first mode shape, (b) the second mode shape, (c) the third mode shape, (d) the fourth mode shape.

The displacement of any point in the structure can be obtained. For example, the displacement of a point along GC is given by

$$y = [1 \quad 1 \quad 0]\mathbf{f}(x)\mathbf{g}_{12}^{+} + [1 \quad 1 \quad 0]\mathbf{f}(-x)\mathbf{g}_{12}^{-} \tag{59}$$

$$u = [0 \quad 0 \quad 1]\mathbf{f}(x)\mathbf{g}_{12}^{+} + [0 \quad 0 \quad 1]\mathbf{f}(-x)\mathbf{g}_{12}^{-} \tag{60}$$

Similarly, the displacement of a point along CB is

$$y = [1 \quad 1 \quad 0]\mathbf{f}(x)\mathbf{c}_{3}^{+} + [1 \quad 1 \quad 0]\mathbf{f}(-x)\mathbf{c}_{3}^{-} \tag{61}$$

$$u = [0 \quad 0 \quad 1]\mathbf{f}(x)\mathbf{c}_{3}^{+} + [0 \quad 0 \quad 1]\mathbf{f}(-x)\mathbf{c}_{3}^{-} \tag{62}$$

The physical parameters of the multiple coupled angular beam structures are the same as those in section 4.2, which the angle is 60° and the boundary A is clamped, B and D are both free. A point transverse force $Q = 1$N is applied at the point G. The excitation point is 0.2 m measured from boundary A and the response point is 0.25 m measured from joint C (Ref: 1×10^{-3}m).

Figure 9 shows the forced frequency response using the present method in this paper and the results obtained by FEM (COMSOL Multiphysics software). It can be seen from the Figure 9 that the two results are in good agreement, which also verified the accuracy of the present approach in this paper.

For the vibration analysis of coupled angular beams or other more complex angular beam structures, the present approach can be summarized in a flowchart as shown in Figure 10.

5. Reflection and transmission coefficients analysis

In this section, reflection and transmission coefficient matrices are analyzed. When the incident waves give rise to

reflected and transmitted waves, the relationship between them can be expanded by using reflection and transmission matrices as follows:

$$\begin{aligned}\mathbf{a}^- &= \begin{bmatrix} a_t^- \\ a_n^- \\ a_l^- \end{bmatrix} = \begin{bmatrix} R_{tt} & R_{tn} & R_{tl} \\ R_{nt} & R_{nn} & R_{nl} \\ R_{lt} & R_{ln} & R_{ll} \end{bmatrix} \begin{bmatrix} a_t^+ \\ a_n^+ \\ a_l^+ \end{bmatrix} = \mathbf{r}\mathbf{a}^+ \\ \mathbf{b}^+ &= \begin{bmatrix} b_t^+ \\ b_n^+ \\ b_l^+ \end{bmatrix} = \begin{bmatrix} T_{tt} & T_{tn} & T_{tl} \\ T_{nt} & T_{nn} & T_{nl} \\ T_{lt} & T_{ln} & T_{ll} \end{bmatrix} \begin{bmatrix} a_t^+ \\ a_n^+ \\ a_l^+ \end{bmatrix} = \mathbf{t}\mathbf{a}^+ \end{aligned} \tag{63}$$

The subscript "*t*," "*n*," and "*l*" represent transverse, near-field, and longitudinal wave components, respectively. R_{tt} is a coefficient of the reflection matrix **r**, which represents the ratio of the reflected transverse wave to the incident transverse wave, that is, the contribution of the incident transverse wave to the reflected transverse wave. Similarly, T_{tt} is the ratio of the transmitted transverse wave to the incident transverse wave. And all the other coefficients can be interpreted in the same way.

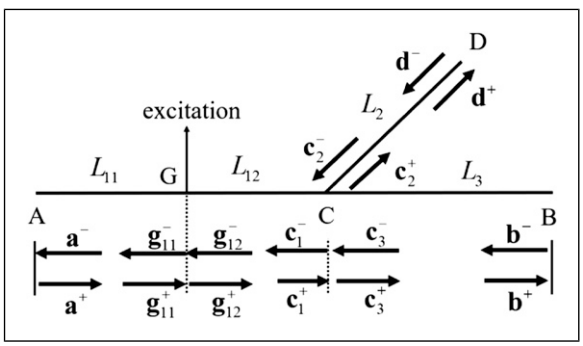

Figure 8. Schematic illustration of forced vibration wave analysis.

5.1. Effects of boundary stiffness

When the size and material of the multiple coupled angular beam structures are determined, the boundary reflection coefficient is only related to the boundary stiffness and external excitation frequency. Boundary spring stiffness is greater than 1×10^{10} N/m (Nm/rad), which can be considered as a clamped boundary. In the present paper, when spring stiffness is 1×10^8 N/m (Nm/rad), there is little error between the calculated results and those of the clamped boundary. So, it can be approximated that $K_L = K_T = 1 \times 10^8$ N/m, $K_R = 1 \times 10^8$ Nm/rad is a clamped boundary.

At the boundary, the longitudinal wave and transverse wave do not interfere with each other. Therefore, the longitudinal wave reflection coefficients are analyzed first. From equations (14) and (63), the longitudinal wave reflection coefficient matrix can be simplified to analyze only the component R_{ll}:

$$a_l^- = R_{ll} a_l^+ = (i + K_L)^{-1}(i - K_L) a_l^+ \tag{64}$$

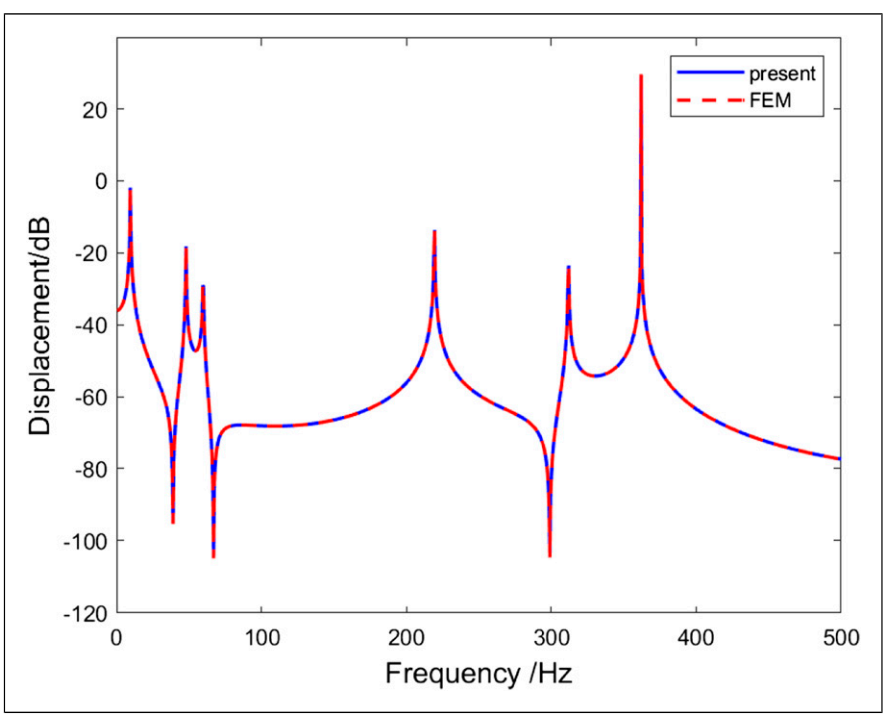

Figure 9. Frequency response of the beam structure subjected to point transverse force.

Through the analysis, for any longitudinal spring stiffness K_L, the coefficient R_{ll} is always unchanged and equal to 1 with the variation of frequency. When $K_L = 0$, the boundary is regarded as a free boundary, and the longitudinal reflected and incident wave have the same amplitude and direction, that is, the displacement at the free boundary is twice that of the incident wave. When $K_L = 1 \times 10^8$ N/m, the boundary is regarded as a fixed boundary, and the longitudinal reflected and incident wave have the same amplitude and opposite direction. And the 2 waves cancel each other, that is, the displacement at the fixed boundary is zero. These can be explained by reflection coefficient R_{ll}.

From equations (14) and (63), the transverse wave reflection matrix can also be simplified as:

$$\begin{bmatrix} a_t^- \\ a_n^- \end{bmatrix} = \begin{bmatrix} R_{tt} & R_{tn} \\ R_{nt} & R_{nn} \end{bmatrix} \begin{bmatrix} a_t^+ \\ a_n^+ \end{bmatrix} = \begin{bmatrix} 1 - iK_T & i(1-K_T) \\ -1 + iK_R & 1 + K_R \end{bmatrix}^{-1} \times \begin{bmatrix} 1 + iK_T & i(1+K_T) \\ 1 + iK_R & -1 + K_R \end{bmatrix} \begin{bmatrix} a_t^+ \\ a_n^+ \end{bmatrix} \tag{65}$$

It can be seen from equation (65) that the transverse wave reflection coefficients are affected by both translational and rotational springs. Figure 11 shows the transverse reflection coefficients as a function of frequency when $K_R = 0$ and K_T varies from 0 to 1×10^8 N/m.

With the increase of frequency, the transverse reflection coefficient R_{tt} is constant and equal to 1 for any translational spring stiffness. The reflection coefficient

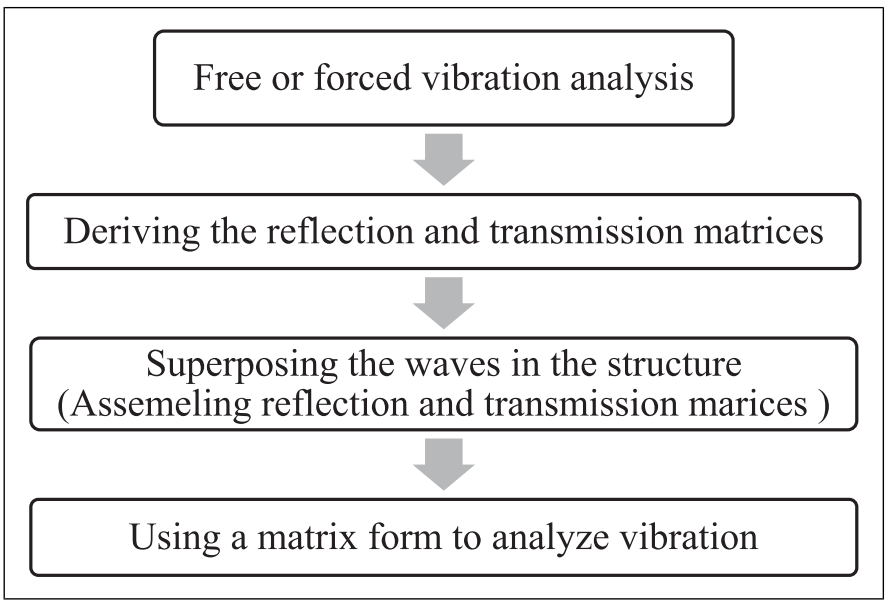

Figure 10. The flowchart of vibration analysis with the present approach.

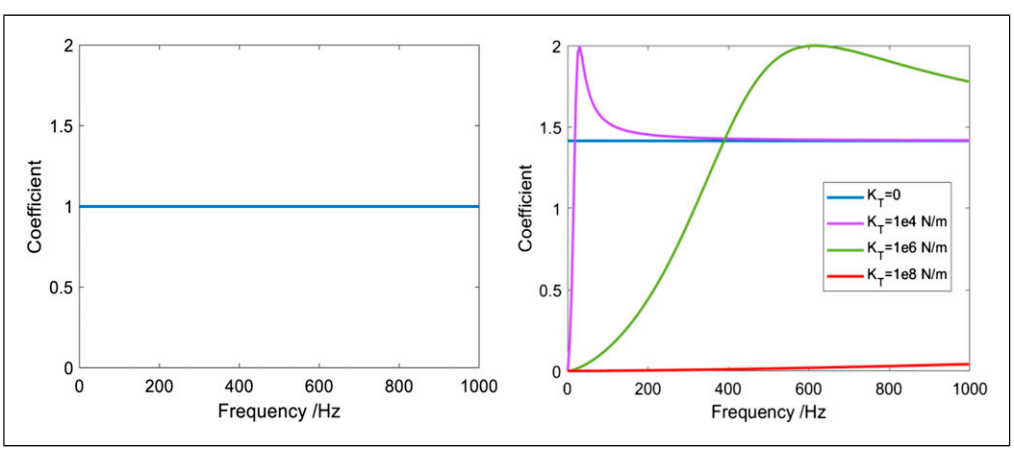

Figure 11. Transverse reflection coefficient as a function of frequency. (a) R_{tt} (b)R_{tn}.

R_{tn} rises from 0 to the maximum value 2, and then gradually decreases and converges to $\sqrt{2}$. The spring stiffness affects the speed of reflection coefficient changing. The greater the stiffness of the spring, the slower the reflection coefficient change. The magnitude and variation of reflection coefficient R_{nt} are the same as those of reflection coefficient R_{tn}. The variation rule of reflection coefficient R_{nn} is similar to that of reflection

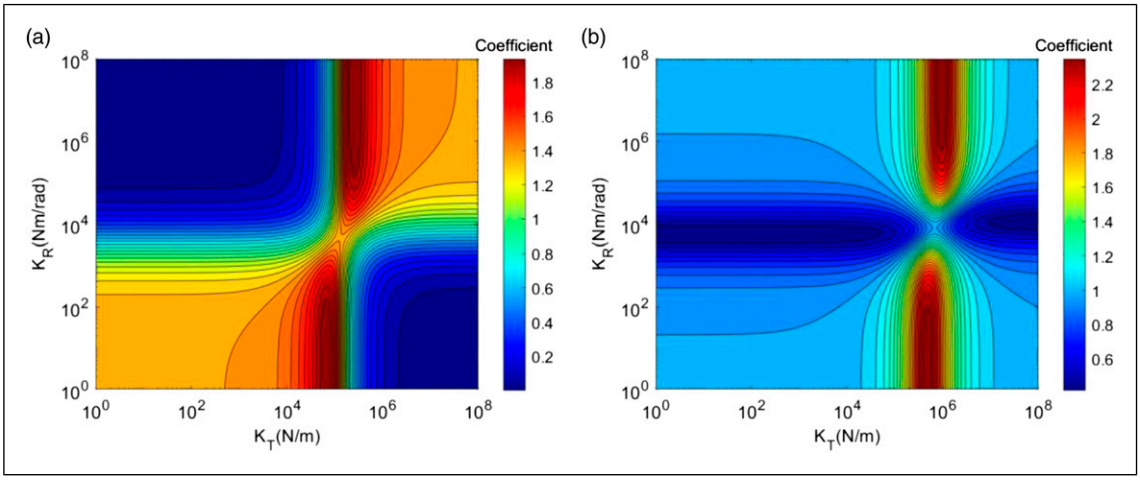

Figure 12. Reflection coefficients as a function of spring stiffness. Freq. = 200 Hz (a) R_{tn} (b)R_{nn}.

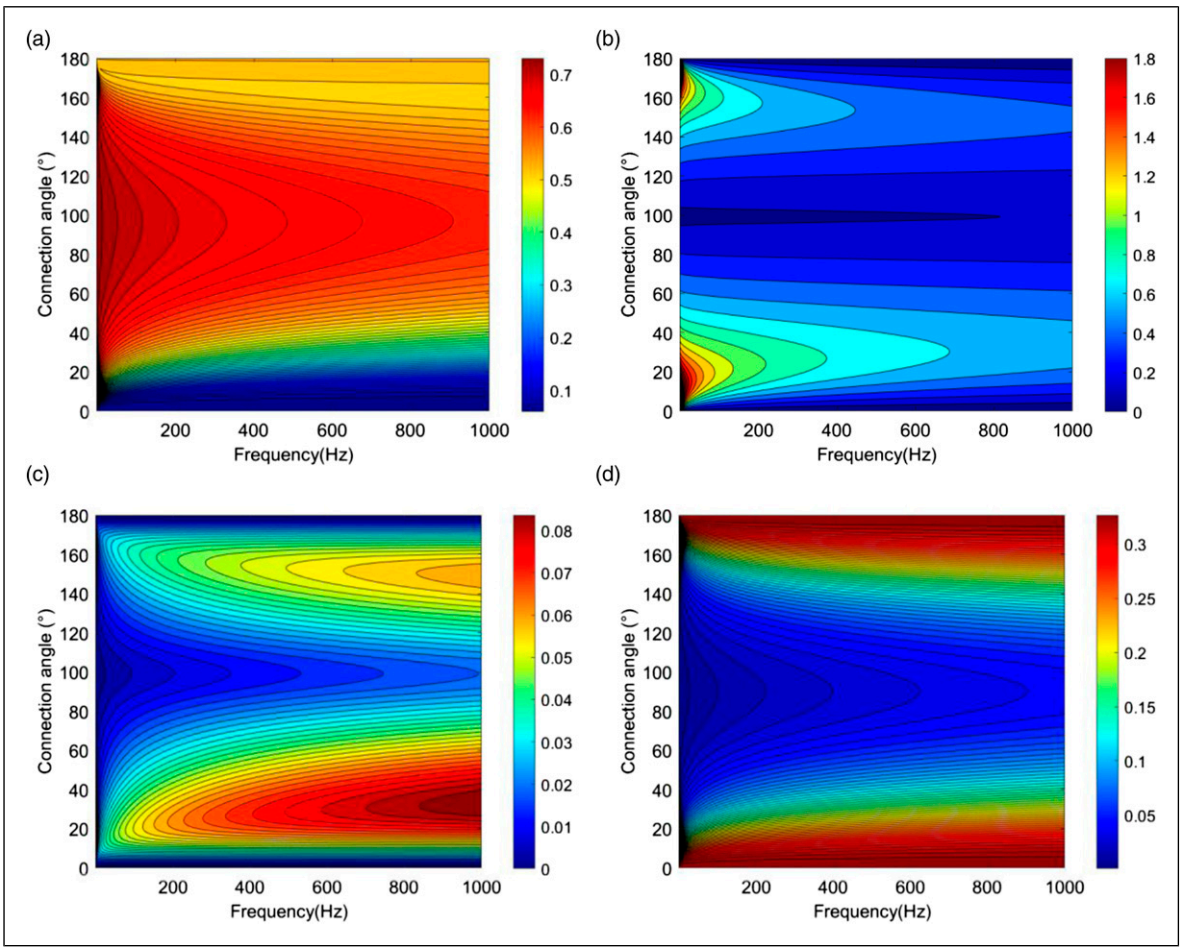

Figure 13. The contour plot of reflection coefficients of beam 1 to beam 1. (a) R_{tt} (b)R_{tl} (c) R_{lt} (d)R_{ll}.

coefficient R_{tn}. However, the reflection coefficient R_{nn} will rise from 1 to about 2.4 first, and then gradually decrease and converge to 1.

Following a similar procedure, the cases that how the reflection coefficient changes with the frequency when K_T and K_R take other different values can also be investigated. Due to the space limitation of this paper, other all cases are not repeated here.

Through the investigation, it is found that no matter how spring stiffness K_T and K_R change, the reflection coefficient R_{tt} is always unchanged and equal to 1. The magnitude and variation of reflection coefficient R_{nt} and R_{tn} are the same. The variation rule of reflection coefficient R_{nn} is similar to that of reflection coefficient R_{tn}. They will vary from a value and converge to another value eventually.

The above content investigates the variation of boundary reflection coefficient with frequency. When the frequency is fixed, the variation of reflection coefficients with spring stiffness will be investigated in the following content. According to the previous analysis, the reflection coefficients R_{ll} and R_{tt} are always unchanged and equal to 1. The magnitude and variation of reflection coefficients R_{tn} and R_{nt} are the same. Therefore, only reflection coefficients R_{tn} and R_{nn} need to be investigated.

Figure 12 shows the reflection coefficient as a function of spring stiffness at 200 Hz. Through the three-dimensional contour plot, it can be known that when the frequency is fixed, the required reflection coefficient can be adjusted by adjusting the boundary spring stiffness. For example, if we want to get the reflection coefficient $R_{tn} = 0.2$, this can be achieved by flexibly adjusting the spring stiffness K_T and K_R according to Figure 12(a). Meanwhile, some numerical data in Figure 11 are listed in Table A1 and Table A2 in Appendix A.

By analyzing the boundary reflection matrix coefficients, it can simplify the calculation and subsequent analysis. If we know the incident wave, we can easily calculate the reflected wave, and then obtain a series of physical quantities such as boundary displacement and velocity, which greatly simplifies the calculation and is also convenient for analysis.

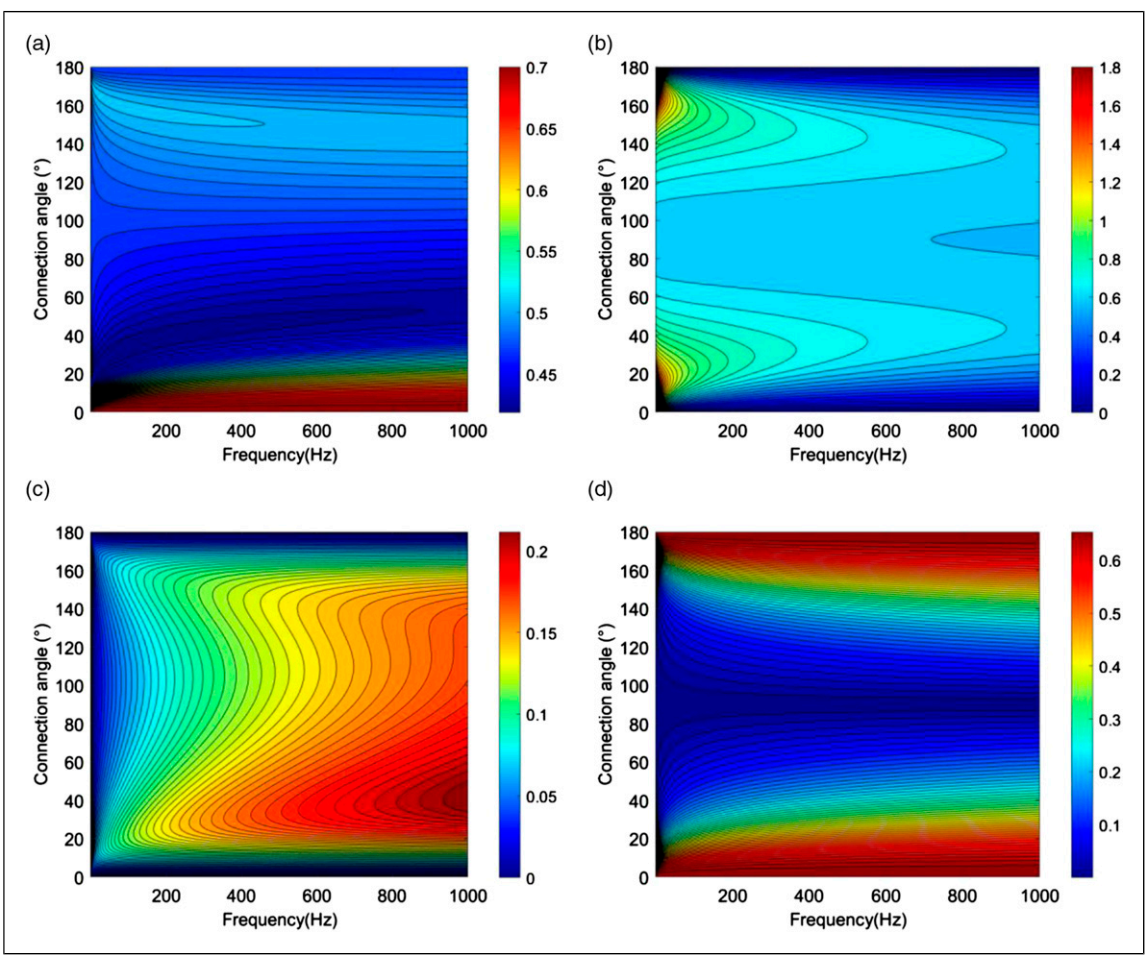

Figure 14. The contour plot of transmission coefficients of beam 1 to beam 2. (a) T_{tt} (b)T_{tl} (c) T_{lt} (d)T_{ll}.

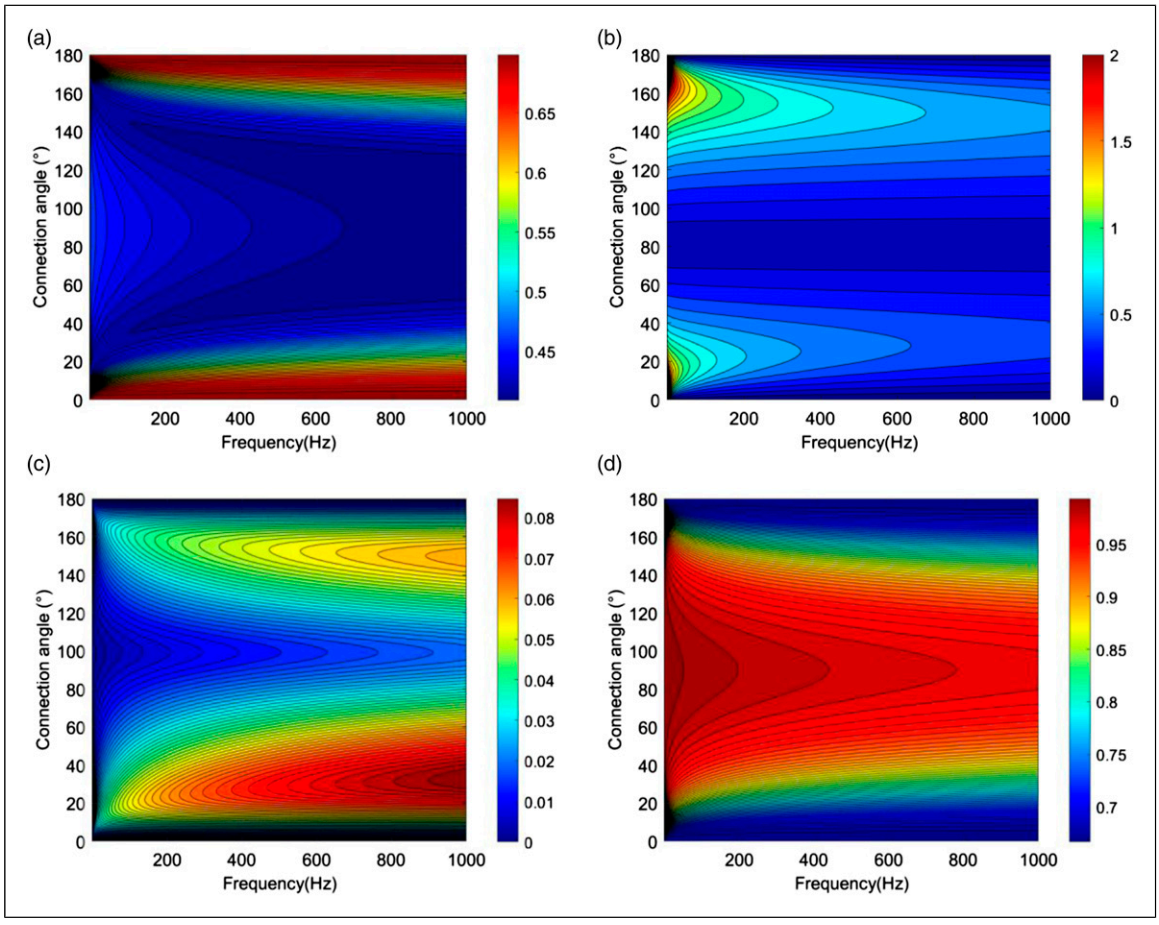

Figure 15. The contour plot of transmission coefficients of beam 1 to beam 3. (a) T_{tt} (b)T_{tl} (c) T_{lt} (d)T_{ll}.

5.2. Effects of joint angle

When the size and material of the multiple coupled angular beam structures are determined, the joint reflection and transmission matrix coefficients are only related to the joint angle and external excitation frequency.

The analysis of the reflection and transmission matrix coefficient of joint is to facilitate the future work of a detailed power flow analysis. The near-filed wave is an attenuated wave, and the energy carried in the far filed is almost zero. The reflected and transmitted energy is only related to the far filed wave. So, the coefficients related to the near-filed wave are not reported. Therefore, only the reflection matrix components R_{tt}, R_{tl}, R_{lt}, and R_{ll} and the transmission matrix components T_{tt}, T_{tl}, T_{lt}, and T_{ll} in equation (62) need to be reported.

Here, taking the model in Figure 4(a) as an example to analyze. The waves $\mathbf{a}^+$ is incident from beam 1, reflected waves $\mathbf{a}^-$ and transmitted waves $\mathbf{b}^+$ propagating to beam 2, and $\mathbf{e}^+$ propagating to beam 3 will be generated at the joint, respectively. The effects of joint angle and frequency on reflection and transmission coefficients are reported in the following Figures 13, Figure 14 and Figure 15. Meanwhile, some numerical data in Figure 13, Figure 14, and Figure 15 are listed in Table A3, Table A4, Table A5, Table A6, Table A7, Table A8, Table A9, Table A10, Table A11, Table A12, Table A13, Table A14, in Appendix A.

As can be seen from Figure 13(a), at the same joint angle, the reflection coefficient R_{tt} gradually decreases with the increase of frequency. At the same frequency, for example, frequency = 400 Hz, the reflection coefficient will gradually increase with the increase of joint angle, reaching the maximum value at about 100°, and then gradually decrease. The desired reflection coefficient can be obtained by adjusting the appropriate joint angle in practical engineering application. All the other reflection and transmission coefficients can be analyzed and reported in the same way and will not be repeated here.

6. Conclusion

In this work, a closed-formed wave-based analysis model is established for in-plane vibration of multiple coupled angular beam structures with general boundary conditions. The reflection matrix at the elastic boundary and the reflection and transmission matrices at the structure joint are derived.

The in-plane free and forced vibrations of the beam structure can be analyzed succinctly and systematically by assembling these reflection and transmission matrices. The procedures are presented using some numerical examples, which show good agreement with the results obtained by FEM. The effects of boundary stiffness and joint angle on reflection and transmission matrix coefficients are demonstrated. In this paper, several results of the response and excitation points on the coupled beams are given to demonstrate the accuracy of the proposed method. In the future research, the current model can be further used for the prediction of dynamic response at different observing and excitation points on the beams, and then the parametric analysis can be performed to study the different vibration characteristics of the complex coupled angular beam structures so as to meet the requirements of vibration and noise reduction in practical engineering. The present approach can be applied to the vibration analysis of other more complex angular beam structures in practical engineering. Furthermore, the proposed method in this work can be extended to the vibration/acoustic analysis in the pipe system conveying fluid, the flexible and long aerospace manipulator, etc.

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This work is supported by the National Natural Science Foundation of China (Grant no. 11972125 and 12102101) and Fok Ying Tung Education Foundation (Grant no.161049).

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Appendix A

Some numerical data of the reflection and transmission coefficients

Table A1. Numerical data of reflection coefficient R_{tn}.

		K_T (N/m)				
Frequency@200 Hz		10^0	10^2	10^4	10^6	10^8
K_R (Nm/rad)	10^0	1.4139	1.4150	1.5261	0.1394	0.0015
	10^2	1.3839	1.3850	1.4936	0.1605	0.0227
	10^4	0.3687	0.3695	0.4477	1.0692	0.9239
	10^6	0.0046	0.0054	0.0843	1.6008	1.4095
	10^8	5.3845e−5	8.1341e−4	0.0797	1.6082	1.4160

Table A2. Numerical data of reflection coefficient R_{nn}.

		K_T (N/m)				
Frequency@200 Hz		10^0	10^2	10^4	10^6	10^8
K_R (Nm/rad)	10^0	0.9998	1.0000	1.0297	1.8584	1.0067
	10^2	0.9753	0.9756	1.0045	1.8296	0.9943
	10^4	0.4953	0.4954	0.5055	0.9545	0.4239
	10^6	0.9920	0.9922	1.0068	2.3722	0.9976
	10^8	0.9999	1.0001	1.0148	2.4107	1.0135

Table A3. Numerical data of reflection coefficient R_{tt} of beam 1 to beam 1.

	200 Hz	400 Hz	600 Hz	800 Hz	1000 Hz
30°	0.4777	0.4108	0.3704	0.3418	0.3201
60°	0.6564	0.6234	0.5995	0.5804	0.5643
90°	0.6905	0.6693	0.6536	0.6407	0.6297
120°	0.6766	0.6522	0.6351	0.6216	0.6103
150°	0.5833	0.5541	0.5392	0.5299	0.5235

Table A4. Numerical data of reflection coefficient R_{tl} of beam 1 to beam 1.

	200 Hz	400 Hz	600 Hz	800 Hz	1000 Hz
30°	0.9191	0.7797	0.6955	0.6360	0.5905
60°	0.4877	0.4607	0.4413	0.4258	0.4127
90°	0.1644	0.1635	0.1628	0.1622	0.1617
120°	0.2728	0.2618	0.2540	0.2479	0.2429
150°	0.6366	0.5426	0.4861	0.4463	0.4161

Table A5. Numerical data of reflection coefficient R_{lt} of beam 1 to beam 1.

	200 Hz	400 Hz	600 Hz	800 Hz	1000 Hz
30°	0.0598	0.0718	0.0784	0.0828	0.0859
60°	0.0317	0.0424	0.0497	0.0554	0.0601
90°	0.0107	0.0150	0.0184	0.0211	0.0235
120°	0.0178	0.0241	0.0286	0.0323	0.0353
150°	0.0414	0.0499	0.0548	0.0581	0.0605

Table A6. Numerical data of reflection coefficient R_{ll} of beam 1 to beam 1.

	200 Hz	400 Hz	600 Hz	800 Hz	1000 Hz
30°	0.1373	0.1667	0.1838	0.1956	0.2044
60°	0.0373	0.0507	0.0604	0.0682	0.0747
90°	0.0189	0.0266	0.0324	0.0373	0.0416
120°	0.0373	0.0507	0.0604	0.0682	0.0747
150°	0.1373	0.1667	0.1838	0.1956	0.2044

Table A7. Numerical data of transmission coefficient T_{tt} of beam 1 to beam 2.

	200 Hz	400 Hz	600 Hz	800 Hz	1000 Hz
30°	0.4338	0.4570	0.4750	0.4892	0.5006
60°	0.4401	0.4330	0.4294	0.4273	0.4261
90°	0.4651	0.4626	0.4606	0.4590	0.4576
120°	0.4840	0.4868	0.4882	0.4889	0.4893
150°	0.5078	0.5087	0.5079	0.5067	0.5005

Table A8. Numerical data of transmission coefficient T_{tl} of beam 1 to beam 2.

	200 Hz	400 Hz	600 Hz	800 Hz	1000 Hz
30°	0.8150	0.7169	0.6575	0.6154	0.5832
60°	0.6542	0.6390	0.6279	0.6189	0.6112
90°	0.5928	0.5895	0.5870	0.5849	0.5830
120°	0.6542	0.6390	0.6279	0.6189	0.6112
150°	0.8150	0.7169	0.6575	0.6154	0.5832

Table A9. Numerical data of transmission coefficient T_{lt} of beam 1 to beam 2.

	200 Hz	400 Hz	600 Hz	800 Hz	1000 Hz
30°	0.1366	0.1662	0.1836	0.1957	0.2049
60°	0.1034	0.1397	0.1655	0.1858	0.2027
90°	0.0863	0.1188	0.1425	0.1618	0.1782
120°	00,876	0.1184	0.1402	0.1574	0.1718
150°	0.1027	0.1250	0.1381	0.1472	0.1541

Table A10. Numerical data of transmission coefficient T_{ll} of beam 1 to beam 2.

	200 Hz	400 Hz	600 Hz	800 Hz	1000 Hz
30°	0.2870	0.3477	0.3826	0.4066	0.4244
60°	0.0768	0.1033	0.1218	0.1363	0.1483
90°	5.6379 E^{-18}	8.2399 E^{-18}	3.0296 E^{-17}	3.0654 E^{-17}	3.0296 E^{-17}
120°	0.0768	0.1033	0.1218	0.1363	0.1483
150°	0.2870	0.3477	0.3826	0.4066	0.4244

Table A11. Numerical data of transmission coefficient T_{tt} of beam 1 to beam 3.

	200 Hz	400 Hz	600 Hz	800 Hz	1000 Hz
30°	0.4326	0.4586	0.4785	0.4942	0.5068
60°	0.4248	0.4155	0.4116	0.4101	0.4098
90°	0.4348	0.4242	0.4177	0.4132	0.4101
120°	0.4248	0.4155	0.4116	0.4101	0.4098
150°	0.4326	0.4586	0.4785	0.4942	0.5068

Table A12. Numerical data of transmission coefficient T_{tl} of beam 1 to beam 3.

	200 Hz	400 Hz	600 Hz	800 Hz	1000 Hz
30°	0.6366	0.5426	0.4861	0.4463	0.4161
60°	0.2728	0.2618	0.2540	0.2479	0.2429
90°	0.1644	0.1635	0.1628	0.1622	0.1617
120°	0.4877	0.4607	0.4413	0.4258	0.4127
150°	0.9191	0.7797	0.6955	0.6360	0.5905

Table A13. Numerical data of transmission coefficient T_{lt} of beam 1 to beam 3.

	200 Hz	400 Hz	600 Hz	800 Hz	1000 Hz
30°	0.0598	0.0718	0.0784	0.0828	0.0859
60°	0.0317	0.0424	0.0497	0.0554	0.0601
90°	0.0107	0.0150	0.0184	0.0211	0.0235
120°	0.0178	0.0241	0.0286	0.0323	0.0353
150°	0.0414	0.0499	0.0548	0.0581	0.0605

Table A14. Numerical data of transmission coefficient T_{tt} of beam 1 to beam 3.

	200 Hz	400 Hz	600 Hz	800 Hz	1000 Hz
30°	0.8799	0.8491	0.8306	0.8177	0.8079
60°	0.9719	0.9609	0.9528	0.9462	0.9406
90°	0.9865	0.9810	0.9768	0.9733	0.9702
120°	0.9719	0.9609	0.9528	0.9462	0.9406
150°	0.8799	0.8491	0.8306	0.8177	0.8079

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